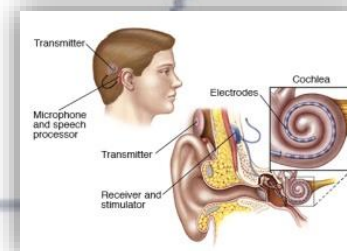
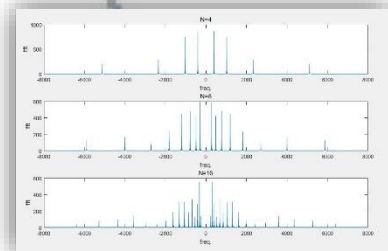
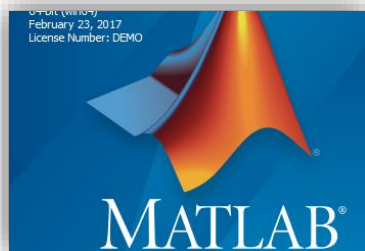


信号与系统实验

主讲人：吴光 博士

Email: wug@sustech.edu.cn



Signals and Systems (Lab)

Lab 3: Fourier Series Representation of Periodic Signals

Part 1

Dr. **Wu Guang**

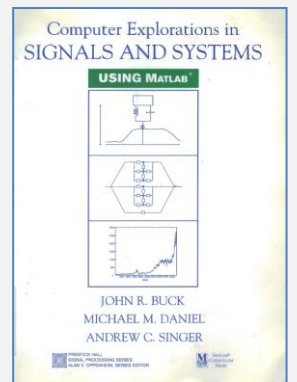
wug@sustech.edu.cn

Electrical & Electronic Engineering
Southern University of Science and Technology

Labs in this course—Review

5 Lab assignments+2 Projects

- Lab 1: MATLAB Programming (3 Weeks)
- **Lab 2: Linear Time-Invariant Systems (2 Weeks)**
- Lab 3: Fourier Series Representation of Periodic Signals (2 Weeks)
- Lab 4: The Continuous-Time Fourier Transform (2 Weeks)
- Lab 5: Coding Test & System, Transform, Convolution and Filter (1 Week)
- Project 1: Speech synthesis and perception with envelope cue (3 Weeks)
- Project 2: Motion detection via communication signals (3 Weeks)



1.1 How to calculate the output of the system ?

- According to the properties of the system, e.g. Linearity, Time-Invariance.
- Step 1: to measure the system function:



- Step 2: to represent the $x(t)$ as a combination of the $\delta(t)/\delta(t - kT_s)$:

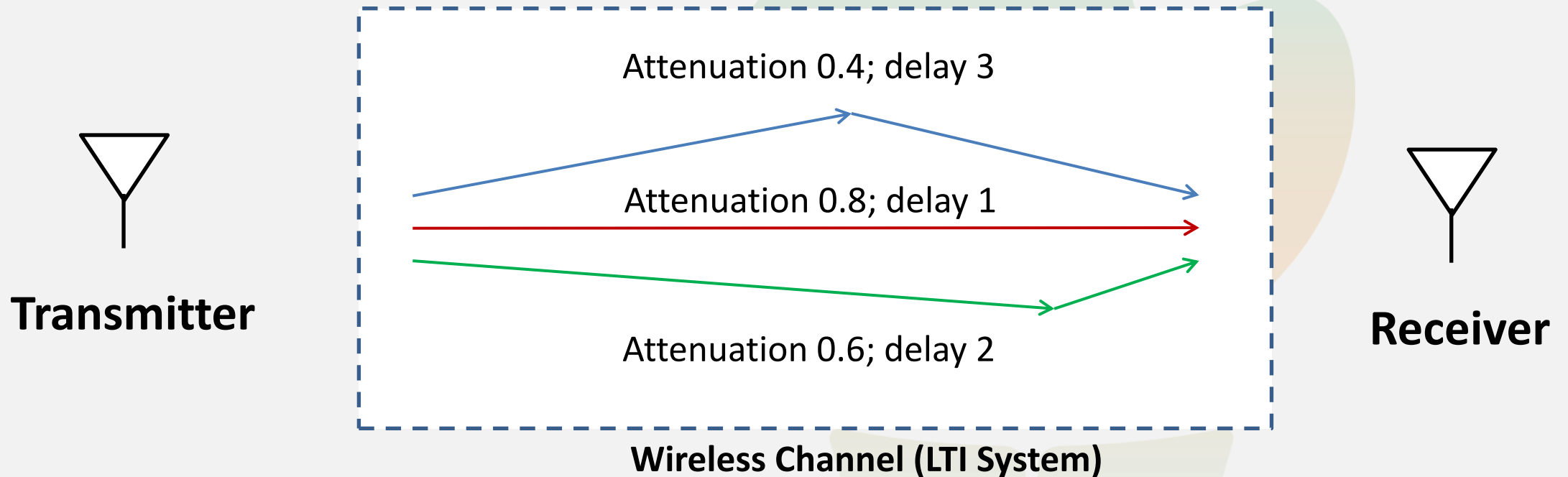


Requirement: H is a LTI system

2.4 conv and filter

	conv	filter
Scenario	• Finite signal • Finite impulse response	• Finite signal • Infinite/finite impulse response
Remarks	• $nw = nu(1) + nv(1) : nu(end) + nv(end)$	• x and y share the same range of time indices • Truncated output

3.1 Application: Simplified Wireless Channel



Impulse response: $h[n] = 0.8\delta[n-1] + 0.6\delta[n-2] + 0.4\delta[n-3]$

Difference equation: $y[n] = 0.8x[n-1] + 0.6x[n-2] + 0.4x[n-3]$

3.2 Application: Echo-Time Estimation

Suppose that you were given $y[n]$ but did not know the value of the echo time, N , or the amplitude of the echo, α . Based on Eq. (2.21), can you determine a method of estimating these values? Hint: Consider the output y of the echo system to be of the form:

$$y[n] = x[n] * (\delta[n] + \alpha\delta[n - N])$$

and consider the signal,

$$R_{yy}[n] = y[n] * y[-n].$$

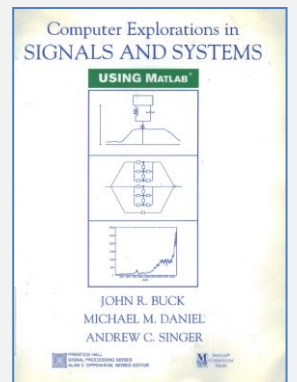
This is called the autocorrelation of the signal $y[n]$ and is often used in applications of echo-time estimation. Write $R_{yy}[n]$ in terms of $R_{xx}[n]$ and also plot $R_{yy}[n]$. You will

$$\begin{aligned} R_{yy}[n] &= x[n] * (\delta[n] + \alpha\delta[n - N]) * x[-n] * (\delta[n] + \alpha\delta[n + N]) \\ &= R_{xx}[n] * ((1 + \alpha^2)\delta[n] + \alpha\delta[n - N] + \alpha\delta[n + N]) \end{aligned}$$

Labs in this course

5 Lab assignments+2 Projects

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Baron Jean Baptiste Joseph Fourier (1768-1830)

Type of Transform	Example Signal
Fourier Transform <i>signals that are continious and aperiodic</i>	
Fourier Series <i>signals that are continious and periodic</i>	
Discrete Time Fourier Transform <i>signals that are discrete and aperiodic</i>	
Discrete Fourier Transform <i>signals that are discrete and periodic</i>	



Baron Jean Baptiste Joseph Fourier
法国数学家、物理学家，1768-1830

1. Complex Exponentials

$$\begin{aligned} x(t) = e^{st} &\longrightarrow \boxed{h(t)} \longrightarrow y(t) = \int_{-\infty}^{+\infty} h(\tau) e^{s(t-\tau)} d\tau \\ &= \left[\int_{-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau \right] e^{st} \\ &= \underbrace{H(s)}_{\text{eigenvalue}} \underbrace{e^{st}}_{\text{eigenfunction}} \end{aligned}$$

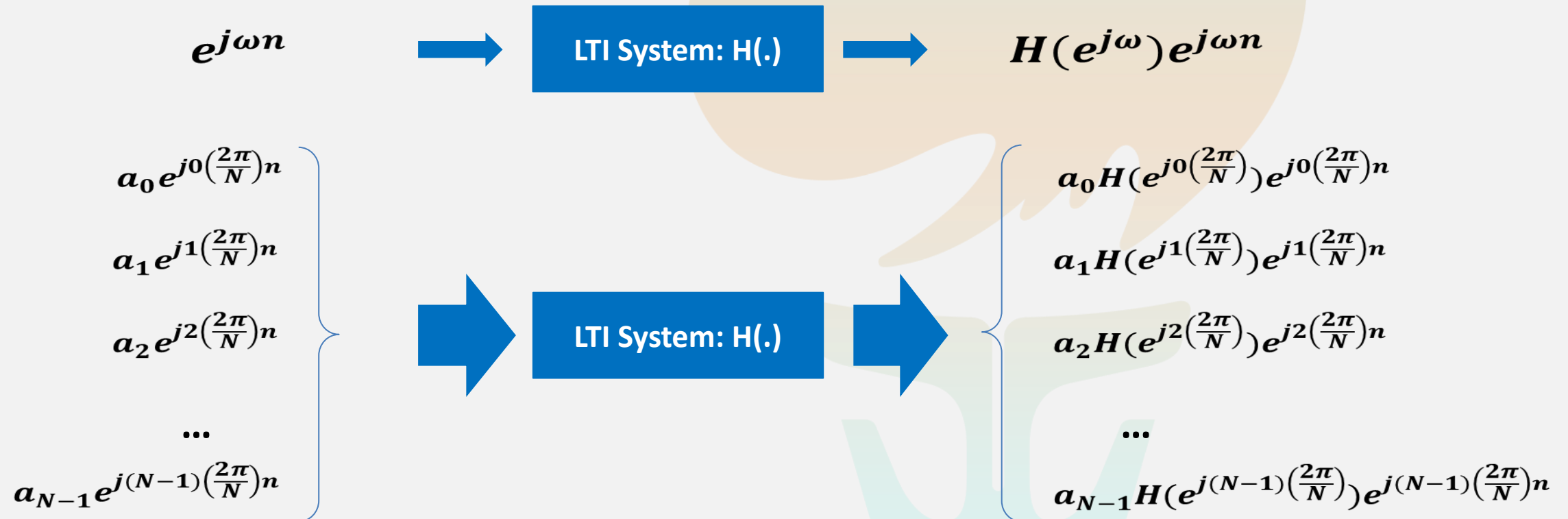
$s = j\omega$ - purely imaginary,
i.e. signals of the form $e^{j\omega t}$



The system H is LTI

2.1 If the system is digital,

➤ We can calculate all the outputs if the system eigenvalue $H(s)$ is known.



2.2 Frequency Response

➤ In this Lab, we will use `freqz()` to calculate Frequency Response :

```
[H omega] = freqz(b, a, N);
```

$$H(e^{j\omega_k}) \quad \omega_k = \left(\frac{\pi}{N}\right)k, 0 \leq k \leq N-1$$

```
[H omega] = freqz(b, a, N, 'whole');
```

$$H(e^{j\omega_k}) \quad \omega_k = \left(\frac{2\pi}{N}\right)k, 0 \leq k \leq N-1$$

```
>> help freqz
```

freqz - Frequency response of digital filter

This MATLAB function returns the n-point frequency response vector `h` and the corresponding angular frequency vector `w` for the digital filter with transfer function coefficients stored in `b` and `a`.

```
[h,w] = freqz(b,a,n)
```

```
[h,w] = freqz(sos,n)
```

```
[h,w] = freqz(d,n)
```

```
[h,w] = freqz(___,n,'whole')
```

```
[h,f] = freqz(___,n,fs)
```

```
[h,f] = freqz(___,n,'whole',fs)
```

```
h = freqz(___,w)
```

```
h = freqz(___,f,fs)
```

```
freqz(___)
```

2.3 Exercise 1: Frequency response

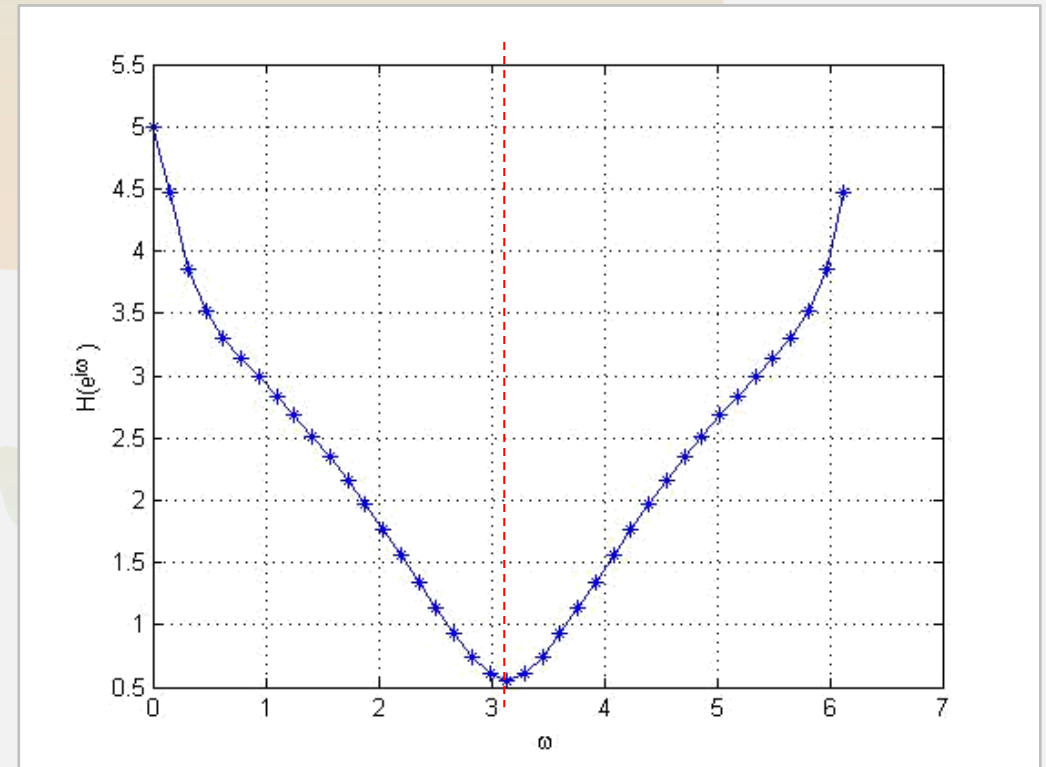
➤ Consider LTI System: $y[n] - 0.8y[n - 1] = 2x[n] - x[n - 2]$

Define the vector of coefficients:

```
A=[1 -0.8];  
B=[2 0 -1];
```

Plot the frequency response:

```
[H Omega] = freqz(B, A, 40, 'whole');  
plot(Omega, abs(H), '*-');  
xlabel('\omega');  
ylabel('H(e^{j\omega})');  
grid;
```



2.4 Exercise 1: Frequency response

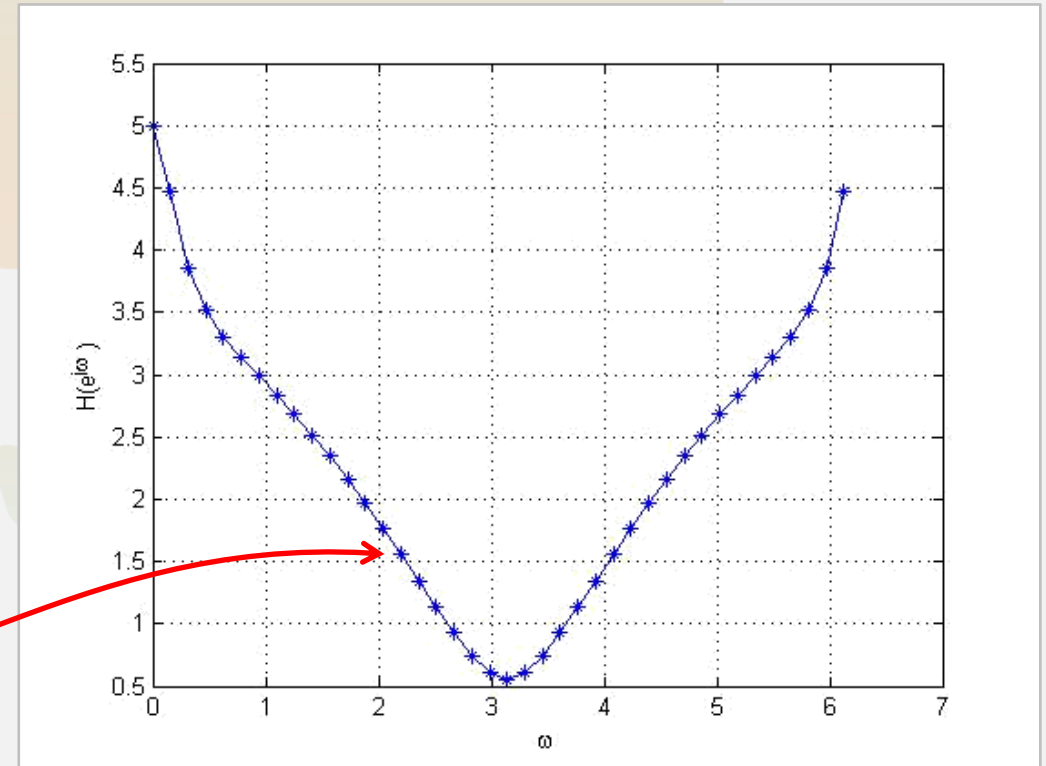
$$a_0 H(e^{j0(\frac{2\pi}{N})}) e^{j0(\frac{2\pi}{N})n}$$

$$a_1 H(e^{j1(\frac{2\pi}{N})}) e^{j1(\frac{2\pi}{N})n}$$

$$a_2 H(e^{j2(\frac{2\pi}{N})}) e^{j2(\frac{2\pi}{N})n}$$

...

$$a_{N-1} H(e^{j(N-1)(\frac{2\pi}{N})}) e^{j(N-1)(\frac{2\pi}{N})n}$$



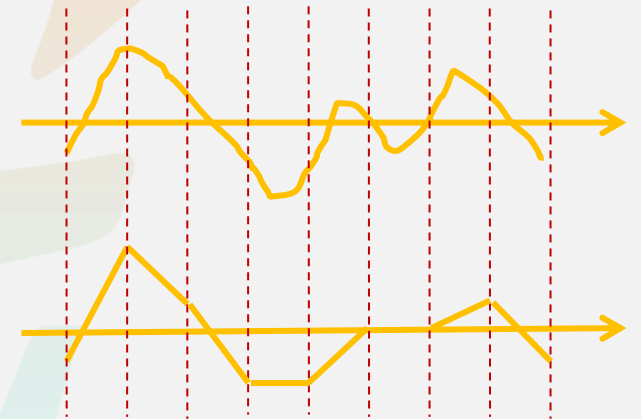
3.1 CT LTI System by Differential Equation

- Reading assignment: textbook 2.4.1.
- Causal CT LTI system can be specified by a linear constant-coefficient differential equation:

$$\sum_{k=0}^K a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^M b_m \frac{d^m x(t)}{dt^m}$$

- Coefficient vectors:

$$\begin{aligned} \mathbf{A} &= [a_K \ a_{K-1} \ \cdots \ a_0] \\ \mathbf{B} &= [b_K \ b_{K-1} \ \cdots \ b_0] \end{aligned}$$



3.2 Attention !

- CT LTI system by differential equation

- $\sum_{k=0}^K a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^M b_m \frac{d^m x(t)}{dt^m}$

- $A = [a_K, a_{K-1}, \dots, a_0]$

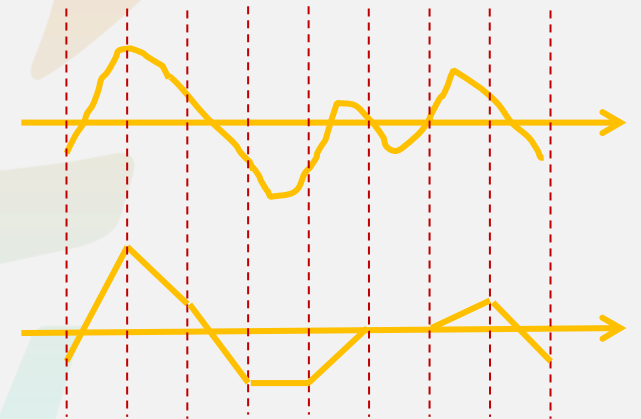
- $B = [b_M, b_{M-1}, \dots, b_0]$

- DT LTI system by difference equation

- $\sum_{k=0}^K a_k y[n - k] = \sum_{m=0}^M b_m x[n - m]$

- $A = [a_0, a_1, \dots, a_K]$

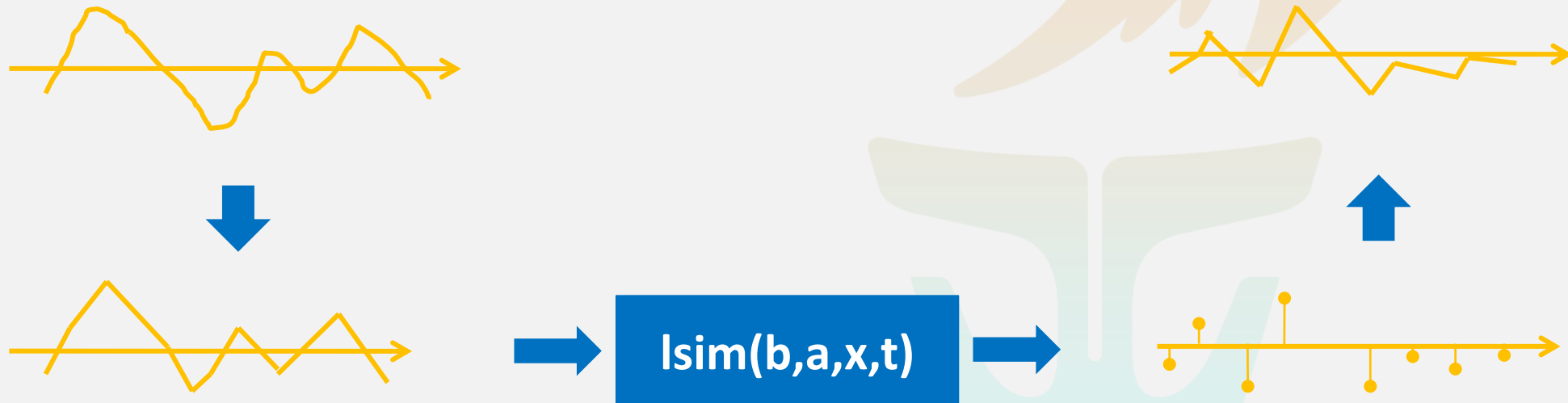
- $B = [b_0, b_1, \dots, b_M]$



3.3 CT LTI System: `lsim`

- **`lsim()`**: generate sampled output according to sampled input signal and CT system function
- **Syntax**: `lsim(b,a,x,t)`

Sampled input signal
Vector of sampling time: `t`
Vector of sampled value: `x`



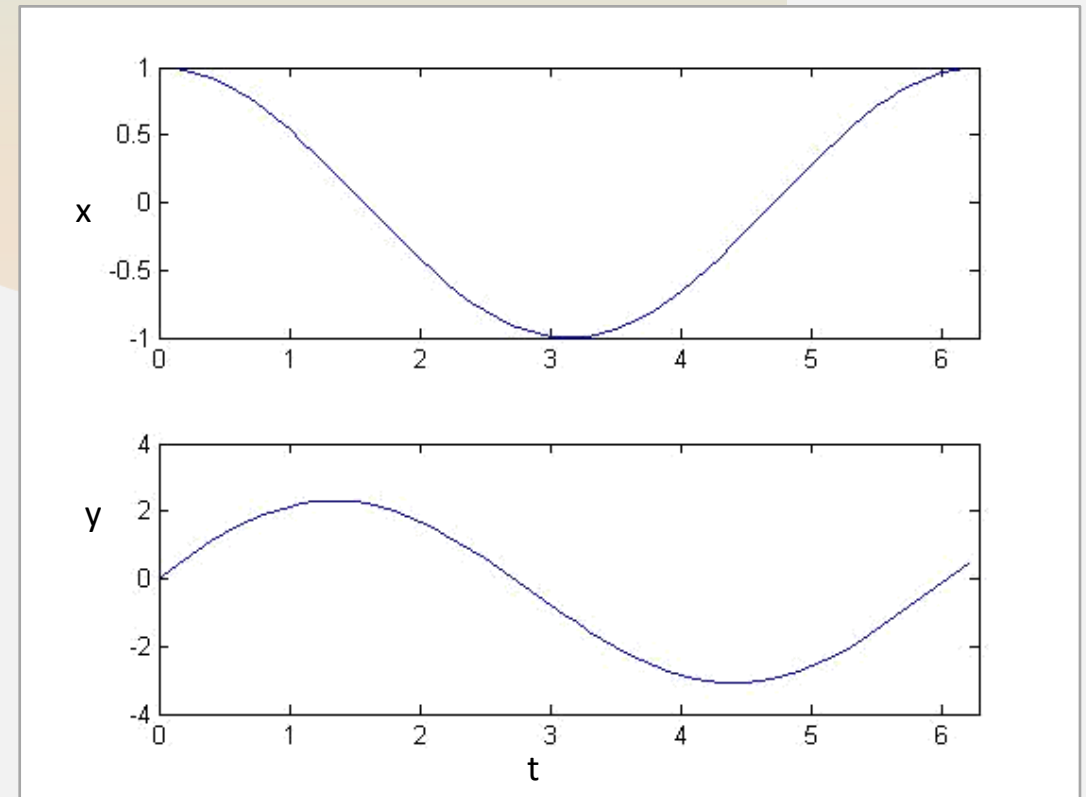
3.4 Exercise 2: CT System

➤ Consider LTI System: $0.3y(t) + dy(t)/dt = 3x(t)$

```
A=[1 0.3];  
B=3;
```

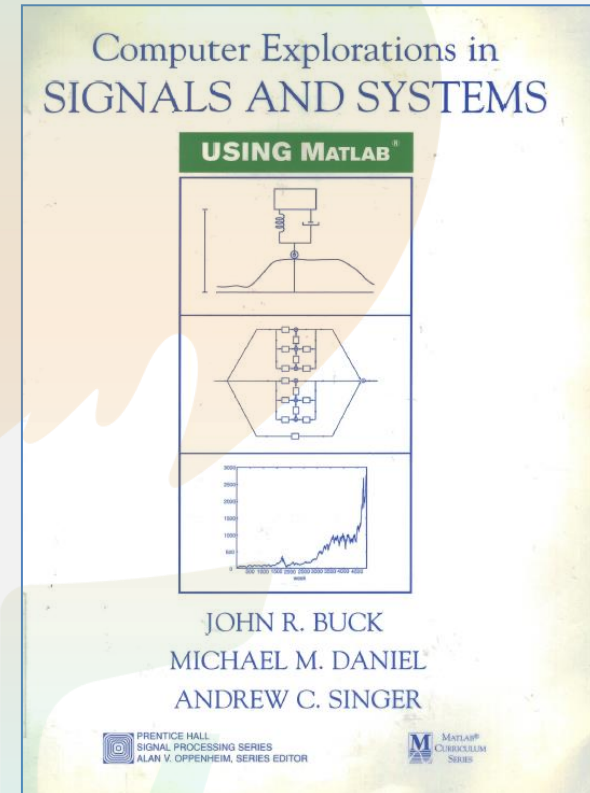
➤ Sample the input signal $x=\cos(t)$:

```
t=0:0.1:2*pi;  
x=cos(t);  
y=lsim(B,A,x,t)';  
subplot(2,1,1), plot(t,x);  
xlim([0 2*pi]);  
subplot(2,1,2), plot(t,y);  
xlim([0 2*pi]);
```



4. Lab Assignment

- Read tutorial 3.2 & 3.3 by yourself
- 3.8、3.10
- Submit your report.



Question ?

